

How to Ensure High Reliability in PCBs for Space Use

This article discusses the critical aspects and characteristics to be kept in mind while designing and manufacturing PCBs destined for use in space products. It also describes the environments that electronic systems face in space

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There is a high level of complexity involved whilst manufacturing PCBs, especially boards destined for military or aeronautical applications like spacecraft. There are certain procedures that have to be followed to create PCBs that work with high reliability in any environment. Let us find out more about these boards, the kind of environments spacecrafts operate in, the main characteristics to be measured while creating such PCBs and, finally, also look at the processes followed at the Indian Space Research Organisation (ISRO) to test them.

Why PCBs for space applications need to be top-notch

About 70 per cent of a spacecraft comprises electronics, and for any electronic device, the PCB is its backbone. It provides mechanical support for the electrical interconnections between components. However, when it comes to space applications, the process of manufacturing the PCB and ensuring its reliability becomes inherently complex.

PCBs targeted for aerospace applications are required to go above and beyond the standard processes and procedures in PCB design, fabrication and assembly. In comparison to previous generations, designs for the military and aerospace sector today demand high reliability. In addition to a 100 per cent fault coverage using techniques such as design for test (DFT), the high RF and the thermal

management require special attention with special PCB design and layout techniques.

System reliability plays an important role in spacecraft. Enormous effort goes into the making of a spacecraft, which is made up of a lot of sub-systems involving huge monetary investments. Besides, the users of the spacecraft have high expectations with respect to its performance, as it

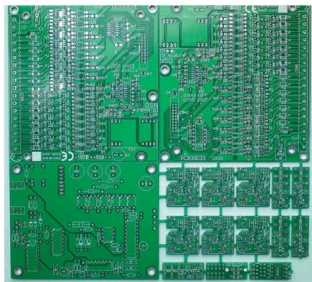
components for consumer PCBs with tin-lead finishes. The components used for spacecraft cost about 5 to 10 times more than the commercial components.

Application-specific PCBs

Spacecrafts incorporate various types of PCBs—single-sided, double-sided and multi-layer boards, rigid flex boards, metal core boards for heat sink

or thermal management purposes, and high-density interconnects for state-of-the-art components. Based on their applications, PCBs are classified into three general types—consumer boards, professional boards and high-reliability boards.

The most common applications are in consumer devices like radios, television sets, cellular phones and even testing equipment. These are



A panel with different types of PCBs (Courtesy: www.madeinchina.com)

can cause human as well as material damage in case of a malfunction while in flight.

The components that populate consumer PCBs have tolerances anywhere between 5 and 10 per cent, whereas military or aeronautical standard components have tight tolerances of 1 to 2 per cent. These components are expensive but they are highly reliable, with gold-finish terminations, compared to

called consumer boards. The materials used to make such boards are less expensive. In this case, the marketing factor is much higher than the reliability factor. It is not so important for the board to have very good and consistent electrical properties, since consumer devices are typically used and later discarded. Professional boards are used for technological developments where the reliability is the most im-